

# Resolving Concurrency and Latency in Agentic LLMs: A Distributed Systems Architecture for Hybrid RAG Memory

Ali Reza  
Department of AI & DS  
Roll No: 23i-0044

Asad Akbar  
Department of AI & DS  
Roll No: 23i-0120

## I. PROBLEM STATEMENT

Advanced RAG architectures enhance Conversational LLMs by separating episodic, semantic, and short-term memory into distinct databases. While this improves accuracy, sequential querying of these stores creates critical latency bottlenecks, preventing real-time deployment. Akbar et al. explicitly identifies this challenge and calls for distributed retrieval strategies.

**Research Question:** How can sequential memory fetch in hybrid RAG be re-architected using PDC techniques to enable concurrent, low-latency retrieval without degrading response quality?

## II. MOTIVATION

Real-time conversational AI requires sub-second responses. Current hybrid RAG architectures, while intelligent, cannot achieve this due to sequential multi-database querying. Parallelizing memory retrieval is essential for practical deployment in customer service, healthcare, and tutoring applications.

## III. OBJECTIVES

- Achieve 3× speedup in memory fetch via parallel database queries.
- Reduce end-to-end latency by 50% versus sequential baseline.
- Implement dynamic load-balancing for heterogeneous database response times.
- Maintain response relevance (Recall@K) versus sequential approach.

## IV. TECHNICAL CONTRIBUTIONS

1. **Parallel Fetch Controller (OpenMP):** Dispatches concurrent non-blocking queries to episodic, semantic, and short-term databases.
2. **Dynamic Load-Balancer:** Monitors per-store latency and adjusts resource allocation in real-time.

3. **Synchronized Fusion:** Aggregates results after all parallel fetches complete (or timeout) for optimal context.

## V. DATASETS & BASELINES

- **Datasets:** Multi-Session Chat, custom long-context dialogues.
- **Baseline:** Sequential Hybrid RAG (Akbar et al. 2026).
- **Comparisons:** Single-document RAG, non-memory LLM.
- **Metrics:** Latency (ms), Speedup (Amdahl’s Law), Parallel Efficiency, Recall@K, Response Relevance.

## VI. METHODOLOGY

- **Phase 1 - Baseline Profiling:** Implement and profile sequential (Episodic → Semantic → Short-Term) pipeline.
- **Phase 2 - Parallel Controller (OpenMP):** Design controller using `#pragma omp task` for concurrent database queries.
- **Phase 3 - Load-Balancing & Sync:** Implement adaptive scheduler and result aggregation barrier.
- **Phase 4 - Evaluation:** Measure latency reduction and response quality versus baseline.

## VII. EXPECTED OUTCOMES

- 3× memory fetch speedup.
- 50% end-to-end latency reduction.
- Preserved response relevance.
- Scalable architecture for real-time conversational AI deployment.

## VIII. Base Reference

Akbar, N. A., Dembani, R., Lenzitti, B., & Tegolo, D. (2025). RAG-driven memory architectures in conversational LLMs—a literature review with insights into emerging agriculture data sharing. *IEEE Access*.